A RESPONSIVE WEBSITE

CSS Grid Layout (Casabona, 2021)

Mobile developments have allowed websites to be viewed on computers, phones, smart watches, or kitchen appliances. It is where the importance of responsive web design comes in. A responsive web design ensures that the website is accessible and visible no matter what device is used.

CSS Grid, or Grid, is a modern solution for creating flexible layouts that work correctly on different screen sizes as it organizes content into two (2) dimensions: rows *and* columns.

Grid uses the `display` property with a second property, such as `grid-template-columns` and `grid-template-rows`. It tells the browser the number of columns to be created and how wide they should be.

```
main {
    display: grid;
    grid-template-columns: 30% 30% 30%;
}
```

This code is used when three (3) equal-width columns are required, with each value specifying each column. The **`main`** tag specifies the main content in the HTML document, and it can enclose several sections/articles but should not contain any repeated content across the document.

This code results in this structure on the webpage:

Figure 1. Retrieved from Casabona, J. (2020). Visual quickstart guide: HTML and CSS. Pearson Education, Inc.

Creating a two-column layout using Grid:

HTML:

```
<main>
    <article>
    ...
    </article>
    <aside>
    ...
    </aside>
</main>
```

Style Sheet:

```
main {
    display: grid;
    grid-template-columns: 68% 28%;
    row-gap: 15px;
    column-gap: 15px;
}
```

The `gap` property can be used to space the rows and columns using `row-gap` and `column-gap` properties.

`<article>` tag specifies independent and self-content and encloses various elements such as header and paragraphs. `<aside>` tag specifies some content besides the content it is placed in and is usually presented as sidebars.

Viewport is the exact area of the screen that is rendering the website. In mobile devices, it is the full screen of a device, while it is the width and height of a browser window.

```html
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
```

This `<meta>` viewport element should be included in all HTML webpages and always go inside the `<head>` element. It gives the browser instructions on controlling the page's dimension and scaling.

The `width=device-width` sets the page width to follow the screen width depending on the device, while `initial-scale=1.0` sets the initial zoom level when the webpage is loaded by the browser.

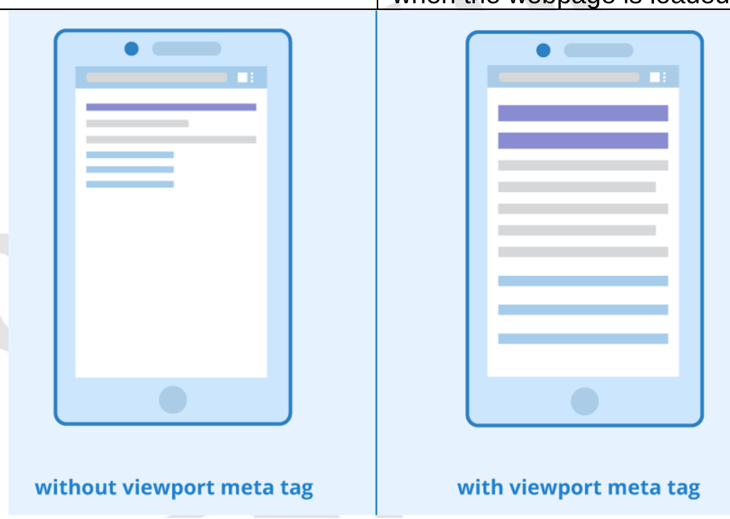

Figure 2. Viewport. Retrieved from https://www.seobility.net/en/wiki/Viewport

MEDIA QUERY (Casabona, 2021)

Media Queries are not just sets of properties and values but containers for other rulesets implemented based on the media query results.

The format for all media queries is as follows:

HTML:

```html
<link rel="stylesheet" media="screen and
(min-width: 1025px)" href="style.css"/>
```

CSS:

```css
@media screen and (min-width: 600px) {
   main {
   display: flex;
   }
}
```

In HTML, the `media` attribute is declared inside a link tag, while it is declared using `@media` in CSS. Note that this affects the grid layout.

and is one of the logics in a media query and is used to combine media types or features. Other logic includes **only** which is used if both conditions are true and **not** which is used to revert the media query that will return false if the output is true.

This code states that if the user is viewing the website on a screen with a viewport of at least 600px wide, it must display all `<main>` elements as `flex`, meaning all flexible items (images, texts) must be the same length. Other media types include all media devices, `print` for printing a webpage, and `speech` for screen readers.

Breakpoint is when the layout changes to adjust to the viewport. It is set using the `min-width`, which makes the layouts more usable and does not break when viewed on any device. For the sample code, it is set to `min-width: 600px`, making 600px the breakpoint.

Responsive Image

A **responsive image** is a flexible one that can be viewed on small devices up to larger screens. The syntax is as follows:

```
<style>
    .responsive {
        width: 100%;
        height: auto;
    }
</style>
```

The `<style>` tags are declared inside the `<head>` tags and below the `<meta>` tag. The selector (`.responsive`) is based on the declared `class` when adding an image such as:

```
<img src="Image.png" alt="alternate text" class="responsive">
```

The given syntax used the CSS `width` property to 100% and `height` to auto, making the image scalable up and down on responsiveness. If the image only needs to be scaled down and never scaled up larger than its original size, `max-width` set to 100% is used.

Responsive Video

A **responsive video** has a video player that can be viewed on small screens up to larger screen devices. The syntax and placement are the same as those used to make responsive images.

```
<style>
    video {
        width: 100%;
        height: auto;
    }
</style>
```

Here, the video player will be responsive and scaled up larger than its original size since the `width` property is 100%. If `max-width` is used and set to 100%, the video player can scale down but never scale up larger than its original size.

References:

Casabona, J. (2021). Visual quickstart guide: HTML and CSS. Pearson Education, Inc.
Donahoe L., Hartl M. (2022). Learn enough HTML, CSS, and layout to be dangerous. Addison-Wesley.